

Predicting Professional Tennis Match Outcomes with Machine Learning

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1. Research Question and Motivation

Our research question is: **Which pre-match factors most strongly influence the outcome of a professional tennis match, and how accurately can we predict the winner using ranking, recent form, court surface, and serve or return statistics?** We chose this topic because all three of us are personally interested in tennis. Additionally, tennis offers a clean prediction setting with a clear outcome, rich historical records, and several variables that are easy to interpret.

This question is interesting because rankings alone do not fully explain match results. Surface type, recent momentum, and serving or returning ability often affect performance in ways that ranking cannot capture by itself. By studying these variables together, we hope to identify which factors matter most and whether a simple predictive model can improve on a ranking-only baseline.

This project is also a strong fit for an introductory data science course because it lets us practice data collection, cleaning, feature engineering, visualization, and predictive modeling on a topic that is both structured and engaging. Prior tennis analytics often highlights ranking, surface, and serve quality as important predictors, and we want to test those ideas directly with historical data.

2. Data Sources

Our primary data source will be the public ATP tennis dataset maintained by Jeff Sackmann on GitHub. It includes yearly match results, player information, rankings, and match-level statistics. We plan to focus on ATP singles matches from about 2010 to 2025 so that we have a large but still modern sample.

If we extend the project, we may also use Jeff Sackmann's Grand Slam point-by-point repository or ATP Tour statistics pages for additional context. However, the main analysis can be completed using the ATP match-level data alone.

We will obtain the data by downloading the relevant CSV files and combining yearly records into one dataset. We will then filter to completed singles matches, standardize dates and player names or IDs, remove or document missing values, and keep variables relevant to our research question, including surface, round, ranking, age, and available match statistics. We also plan to create pre-match features such as ranking gap, age gap, recent win rate, and recent win rate on the same surface.

We expect several preprocessing challenges. Some older matches have missing statistics, and the original files are often organized from the winner's perspective, which creates a risk of target leakage. We will address this by removing incomplete matches when necessary, documenting missing-data decisions, and making sure that all predictors are based only on information available before the match.

Since this dataset is publicly available and licensed for non-commercial use with attribution, it appears appropriate for an academic project as long as we cite it properly and follow any course-specific rules.

3. Methodology and Analysis Plan

We will begin with exploratory data analysis to understand the structure of the dataset and identify useful patterns. This part will include summary statistics and visualizations showing how outcomes vary by ranking difference, court surface, and selected serve statistics. These results will guide our later feature selection.

Our first benchmark will be a simple baseline that predicts the higher-ranked player to win. We will then build a logistic regression model using pre-match variables such as ranking gap, surface, age difference, and recent form. After that, we plan to compare it with at least one tree-based method such as a decision tree or random forest. If time permits, we may also test a more flexible model such as gradient boosting.

We will evaluate our models using a time-based train-test split so that earlier seasons are used for training and later seasons are used for testing. This setup better reflects a realistic prediction task and reduces leakage across time. Our main evaluation metrics will be accuracy, F1 score, ROC-AUC, and either log loss or the Brier score.

We will use Python and Jupyter Notebook for the full workflow. Pandas and NumPy will be used for data cleaning, Matplotlib and Seaborn for visualization, and Scikit-learn for modeling. If needed, we may also use Statsmodels for additional statistical interpretation.

4. Expected Outcomes and Evaluation

We expect ranking difference to be one of the strongest predictors of match outcomes, but we also expect surface type, recent form, and serve or return performance to add useful predictive power. One likely result is that ranking alone misses surface-specific strengths, such as players who perform much better on clay or grass than their overall ranking suggests.

We will consider the project successful if our model performs better than the ranking-only baseline and if the analysis produces clear, interpretable conclusions about what drives tennis outcomes. Even a modest improvement in prediction would be meaningful if it shows which variables consistently provide extra information.

If we have time to extend the analysis, we could compare ATP and WTA matches, build separate models for each surface, or study upset matches in which lower-ranked players defeat higher-ranked opponents. These extensions would broaden the project while staying connected to the same data science workflow.